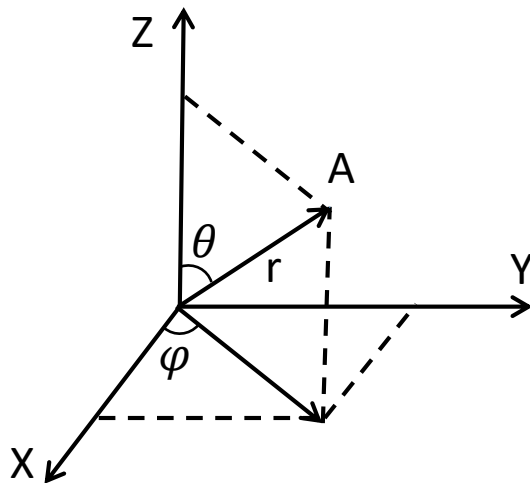


## Cartesian and Spherical coordinates

$$A = (x, y, z) \text{ or } (\theta, \varphi, r)$$

$$\begin{cases} x = r \cdot \sin(\theta) \cos(\varphi) \\ y = r \cdot \sin(\theta) \sin(\varphi) \\ z = r \cdot \cos(\theta) \end{cases}$$



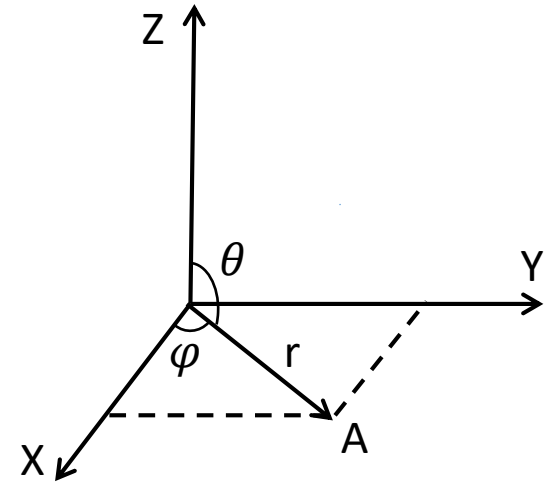
$$\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \cos^{-1}(z/r) \\ \varphi = \tan^{-1}(y/x) \end{cases}$$

1.

The point A has spherical coordinates  $\theta = \pi/2$ ,  $\phi = \pi/3$ ,  $r = 6$ ; the Cartesian coordinates of the point B are  $x = 5$ ,  $y = 5\sqrt{3}$ ,  $z = 0$ . Find the distance between the points A and B.

What are the Cartesian coordinates of the point A, if the Spherical coordinates are  $\begin{pmatrix} \theta = \pi/2 \\ \phi = \pi/3 \\ r = 6 \end{pmatrix}$ ?

$$\begin{cases} x = r \cdot \sin(\theta) \cos(\phi) = 3 \\ y = r \cdot \sin(\theta) \sin(\phi) = 3\sqrt{3} \\ z = r \cdot \cos(\theta) = 0 \end{cases} \longrightarrow A = (3, 3\sqrt{3}, 0)$$



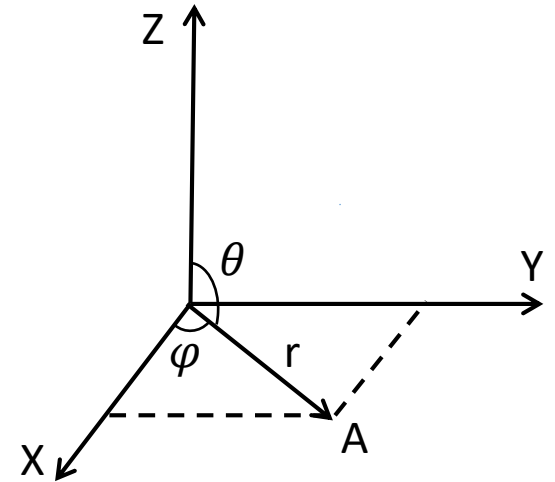
Solution

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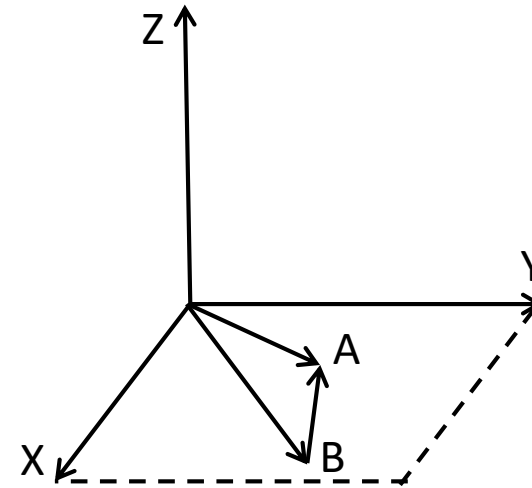
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$$A = (3, 3\sqrt{3}, 0)$$

$$B = (5, 5\sqrt{3}, 0)$$

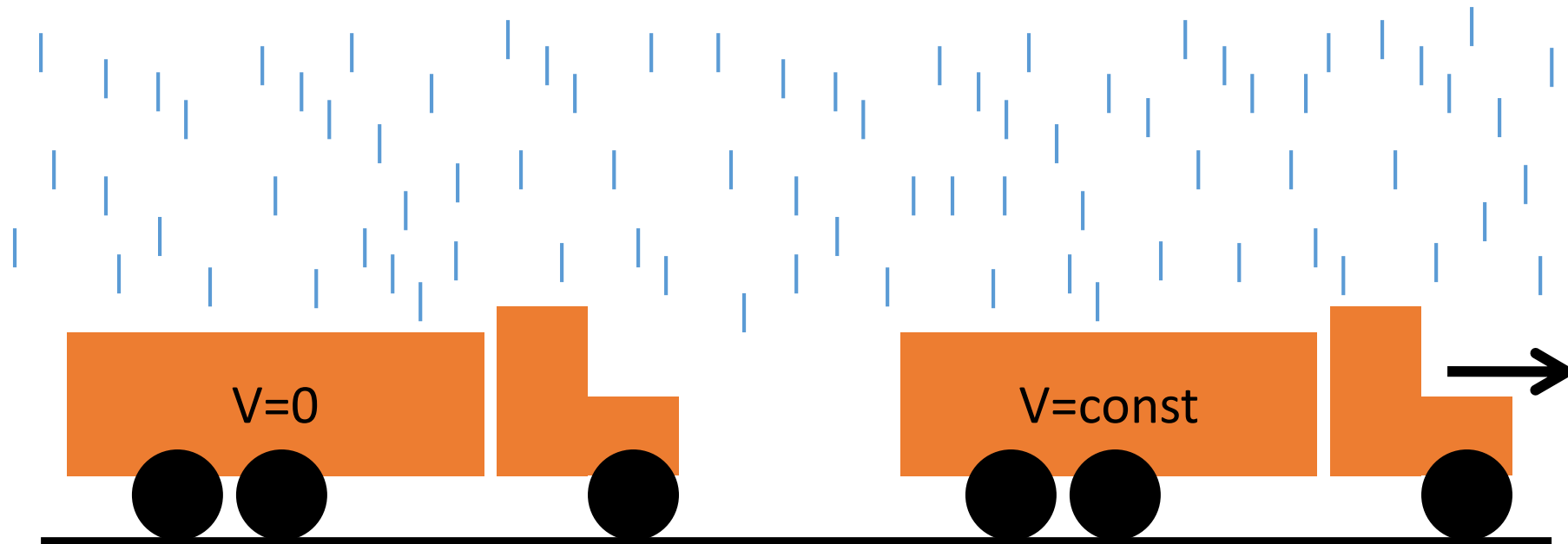


$$|\vec{B} - \vec{A}| = \sqrt{(b_x - a_x)^2 + (b_y - a_y)^2 + (b_z - a_z)^2} = \sqrt{(5 - 3)^2 + (5\sqrt{3} - 3\sqrt{3})^2 + (0 - 0)^2} = 4$$

Solution

2.

Two identical trucks are caught in the rain. First truck is at rest, second one is moving with constant velocity  $V$ . Which truck gets more rain water per minute through the open dump truck body?

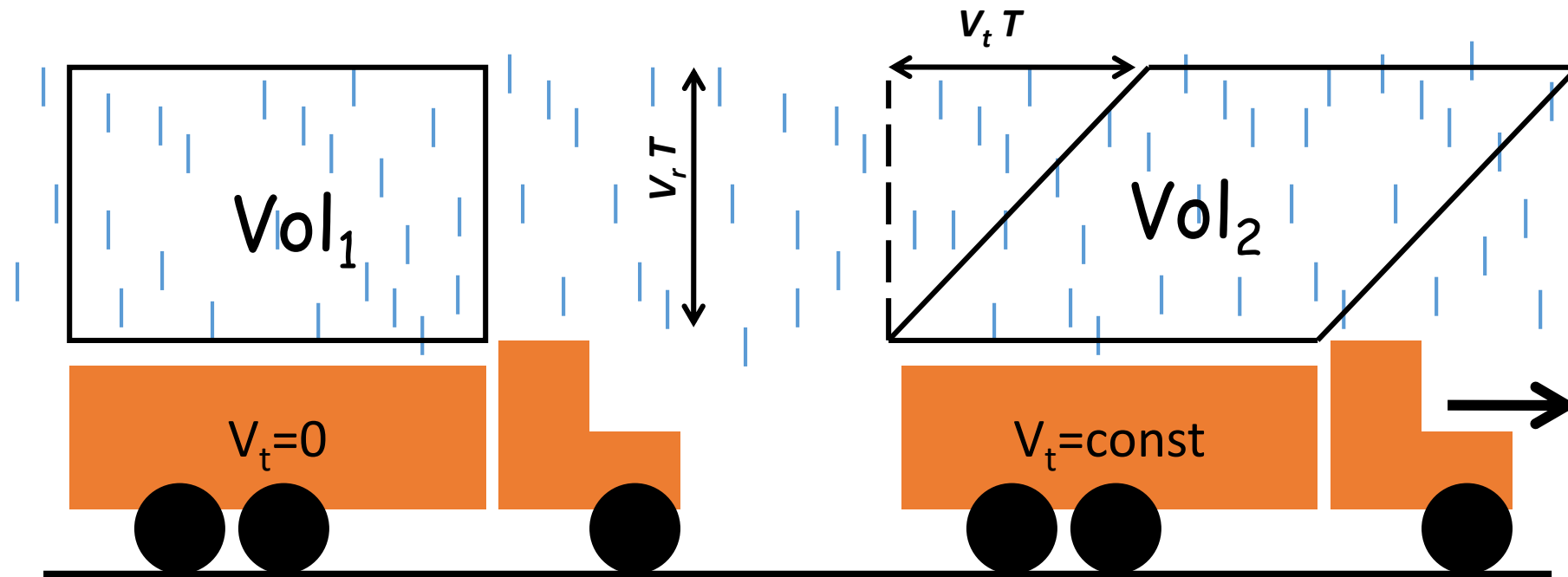


2.

Two identical trucks are caught in the rain. First truck is at rest, second one is moving with constant velocity  $V_t$ . Which truck gets more rain water per minute through the open dump truck body?

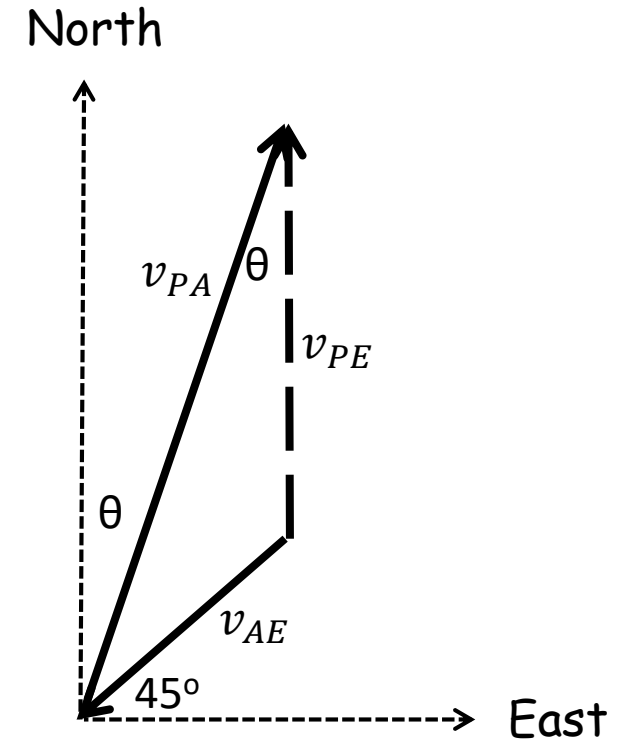
Let's consider the volumes  $Vol_1$  and  $Vol_2$  containing rain drops which get into dump truck bodies during time interval  $T$ .

Graphical solution



Answer:  $Vol_1 = Vol_2$ .

3. A pilot with an airspeed (a speed with respect to air) of  $v_{PA} = 120$  km/h wishes to fly due to north. A wind  $v_{AE} = 40$  km/h is blowing from the northeast. In what direction  $\theta$  should she head, and what will be her ground speed  $v_{PE}$  (a speed with respect to ground)?



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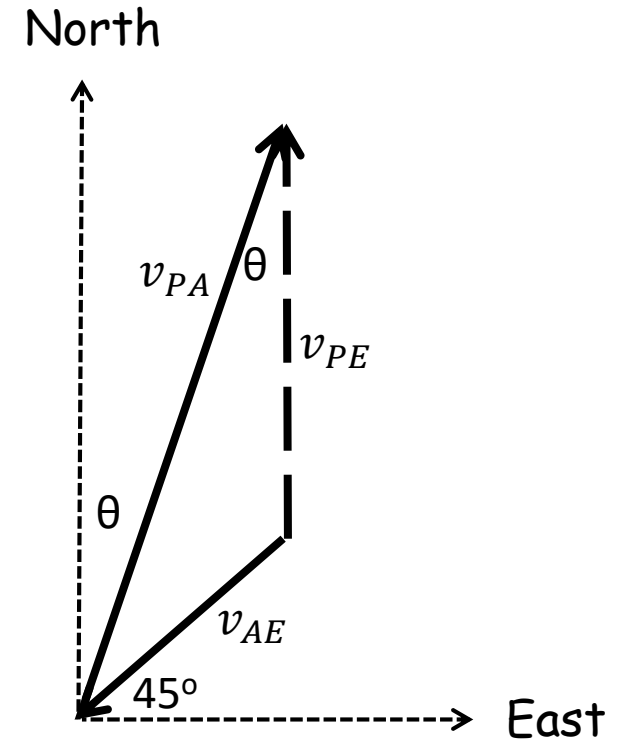
$$(v_{PE} + v_{AE} \cos 45^\circ)^2 + (v_{AE} \sin 45^\circ)^2 = v_{PA}^2$$

$$(v_{PE} + 40 \cos 45^\circ)^2 = (120)^2 - (40 \sin 45^\circ)^2$$

$$v_{PE} = 88.3 \text{ km/h}$$

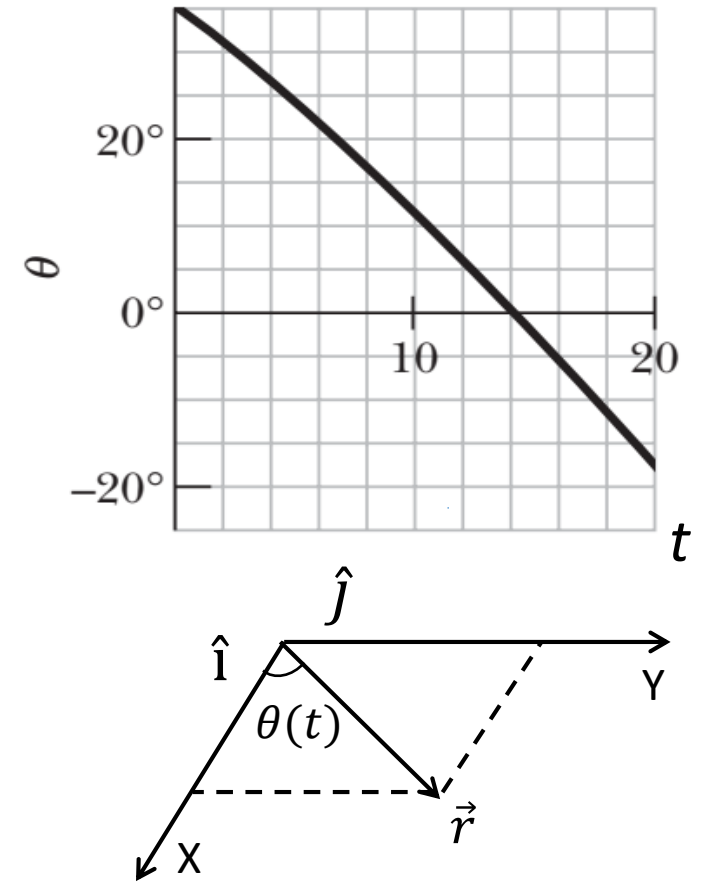
$$\theta = \sin^{-1} \frac{v_{AE} \sin 45^\circ}{v_{PA}} = \sin^{-1} \frac{(40 \sin 45^\circ)}{120} = 13.6^\circ \text{ E of N}$$

The plane should head  $13.6^\circ$  east of north.



4.

The position vector  $\vec{r} = 5t \cdot \hat{i} + (et + ft^2)\hat{j}$  locates a particle as a function of time  $t$ . The vector  $\vec{r}$  is in meters,  $t$  is in seconds. The figure gives the angle  $\theta$  of the position vector's direction as a function of time ( $\theta$  is measured from the positive  $x$  direction). What are the values of the constants  $e$  and  $f$ ?



4.

From vector representation to independent coordinate functions  $x(t)$  and  $y(t)$

$$\vec{r} = 5t \cdot \hat{i} + (et + ft^2)\hat{j} \longleftrightarrow \begin{cases} x(t) = 5 \cdot t \\ y(t) = e \cdot t + f \cdot t^2 \end{cases}$$

Let's build a function connecting  $\theta$  and  $t$  variables

$$\tan(\theta(t)) = \frac{y(t)}{x(t)} = \frac{(e \cdot t + f \cdot t^2)}{(5 \cdot t)} = \frac{(e + f \cdot t)}{5}$$

At two moments  $t = 0$  and  $t = 14$  we have

$$t = 0$$

$$\tan\left(35 \cdot \frac{\pi}{180}\right) = \frac{(e + f \cdot 0)}{5}$$

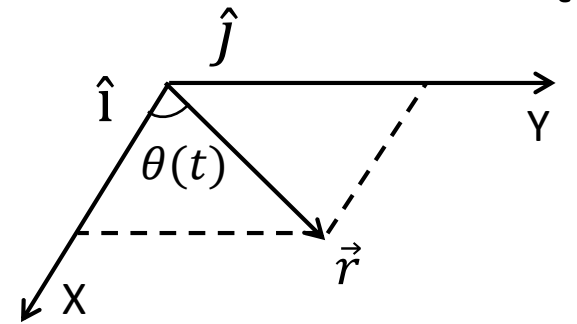
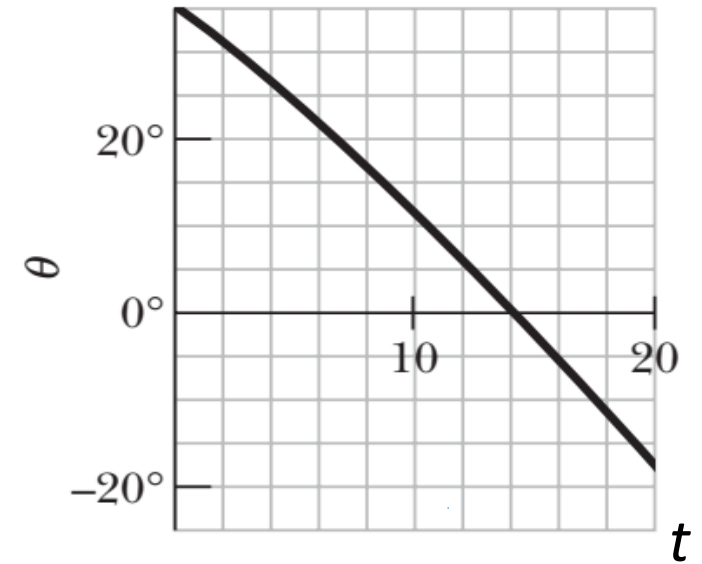
$$0.7 = \frac{e}{5}$$

$$e = 3.5$$

$$t = 14$$

$$\tan(0) = \frac{3.5 + f \cdot 14}{5}$$

$$f = -0.25$$



Answer: 
$$\begin{cases} e = 3.5 \\ f = -0.25 \end{cases}$$

Solution

4.

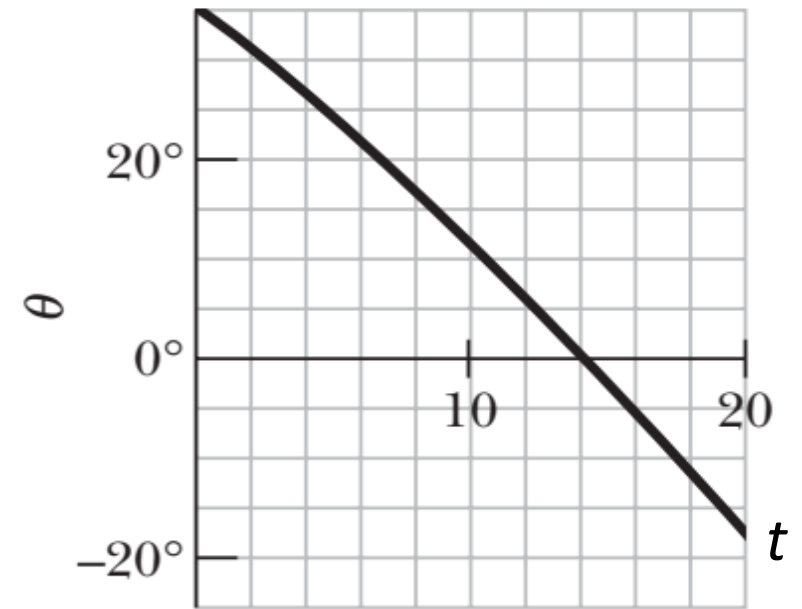
$$\vec{r} = 5t \cdot \hat{i} + (et + ft^2)\hat{j} \quad \longleftrightarrow \quad \begin{cases} x(t) = 5 \cdot t \\ y(t) = e \cdot t + f \cdot t^2 \end{cases}$$

$$\tan(\theta(t)) = \frac{y(t)}{x(t)} = \frac{(e \cdot t + f \cdot t^2)}{(5 \cdot t)} = \frac{(e + f \cdot t)}{5}$$

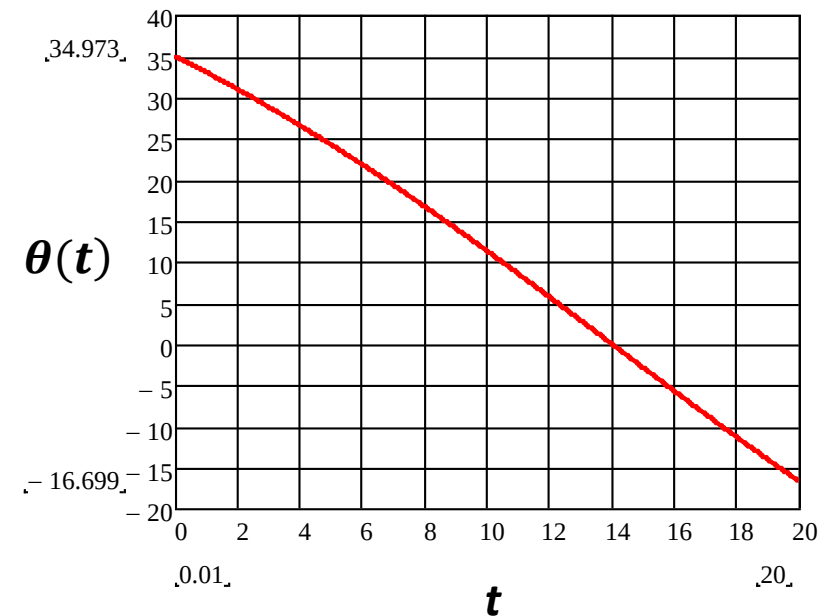
Solution

Checking the solution by building the graph  $\theta(t)$

$$\theta(t) = \text{atan}\left(\frac{3.5t + (-0.25)t^2}{5t}\right)$$

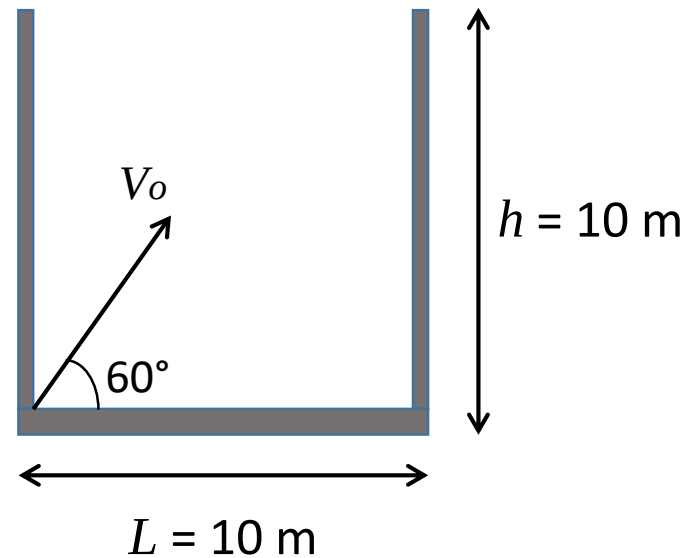


$$e = 3.5, f = -0.25$$



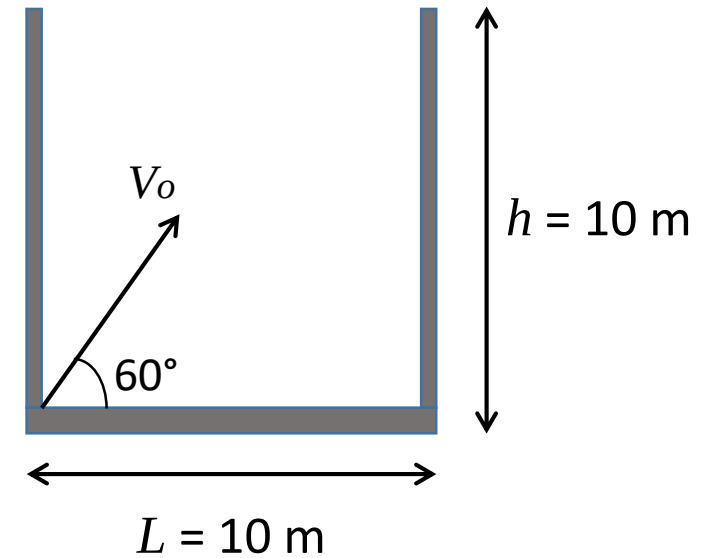
5.

The ball is thrown from the bottom of a rectangular pit with an initial velocity  $V_0 = 20$  m/s which direction makes an angle of  $\alpha = 60^\circ$  to the horizon line. The pit depth is  $h = 10$  m, the distance from the throwing point to the pit wall is  $L = 10$  m. Will the ball leave the pit?



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Solution

Transform from 2-dimensional motion to  $2 \times 1$ -dimensional motions

$$V_x = V_0 \cdot \cos(\alpha)$$

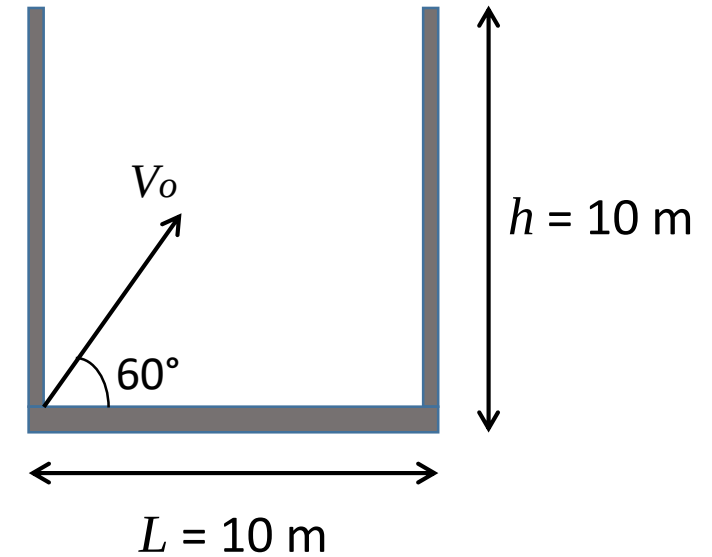
$$V_y = V_0 \cdot \sin(\alpha)$$

time of flying to the wall

$$t = \frac{L}{V_x} = \frac{L}{(V_0 \cdot \cos(\alpha))}$$

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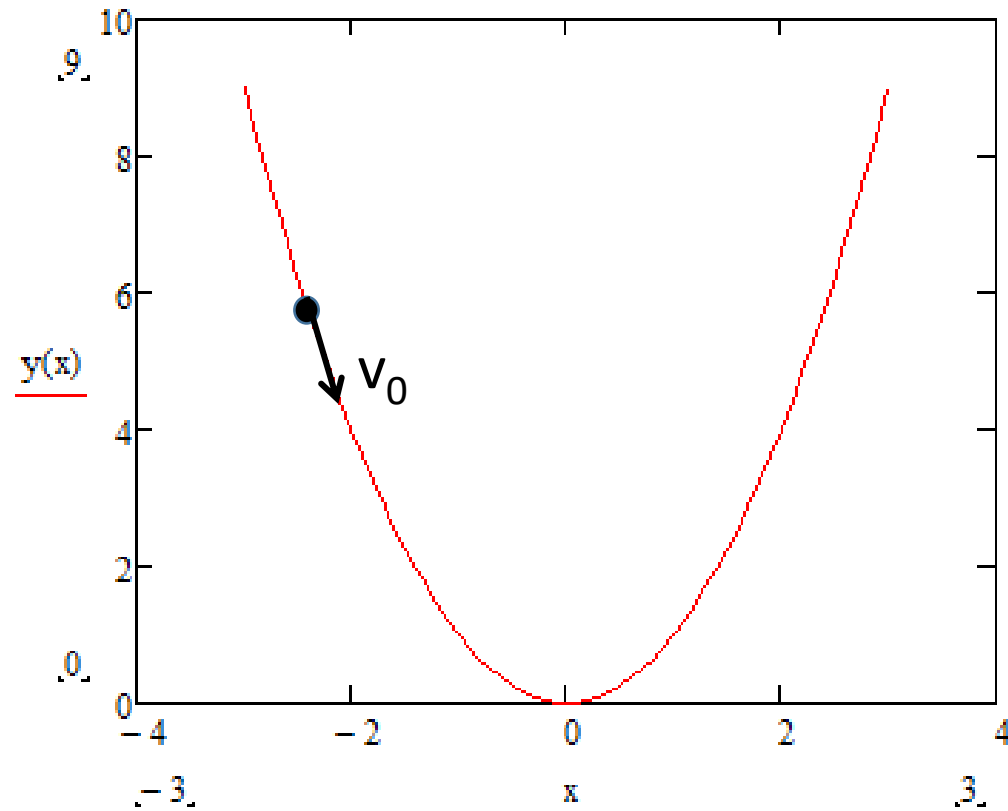
Elevation at moment  $t = \frac{L}{V_x} = \frac{L}{(V_0 \cdot \cos(\alpha))}$

$$H = V_y \cdot t - \frac{g \cdot t^2}{2} = \frac{L \cdot \sin(\alpha)}{\cos(\alpha)} - \frac{g \cdot L^2}{2 \cdot (V_0 \cdot \cos(\alpha))^2} = 12.4 \text{ (m)}$$

Answer:  $H > h$ , the ball will leave the pit,

Solution

6. The material point moves in plane with a constant speed  $v_0$ . Its trajectory is given by equation  $y(x) = c x^2$ . Find at the point  $x = 0$  the acceleration of the material point and the trajectory radius.



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Let's consider the moment  $t$ , when  $x(t) = y(t) = 0$  :

x-motion: velocity and acceleration

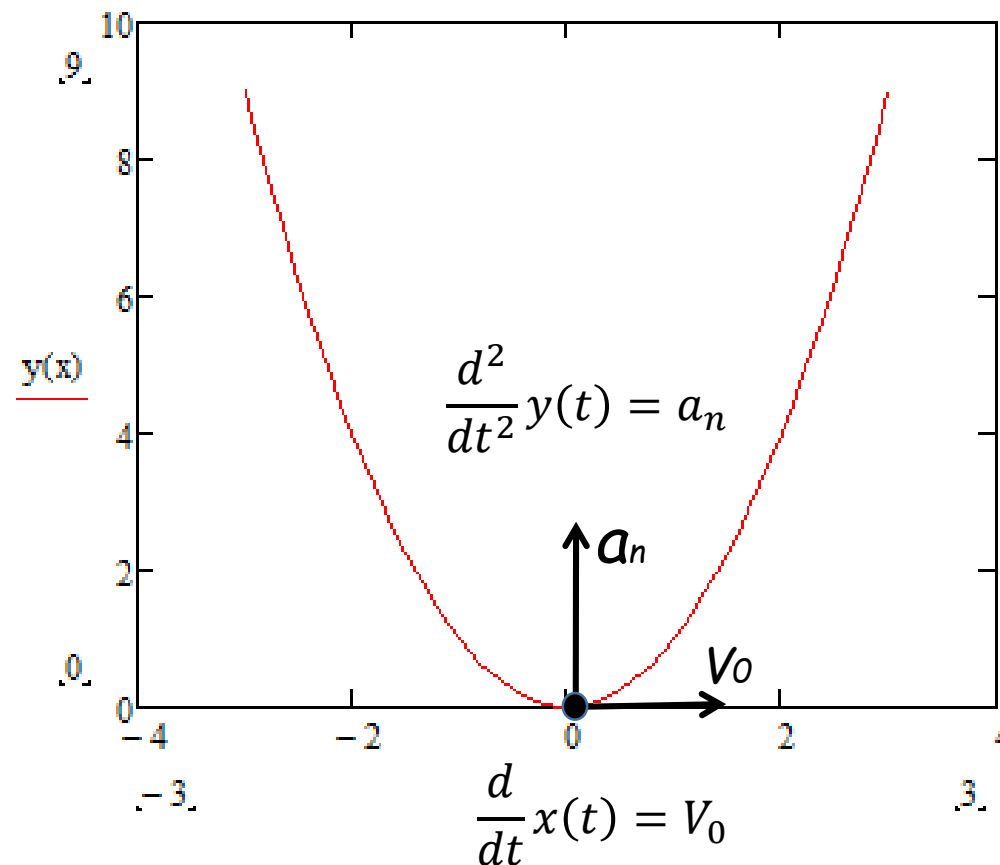
$$\frac{d}{dt}x(t) = V_0$$

$$\frac{d^2}{dt^2}x(t) = 0$$

y-motion: acceleration

$$\frac{d}{dt}y(t) = 0$$

$$\frac{d^2}{dt^2}y(t) = a_y = a_n$$



Solution

6.

The material point moves in plane with a constant speed  $v_0$ . Its trajectory is given by equation  $y(x) = cx^2$ . Find at the point  $x = 0$  the acceleration of the material point and the trajectory radius.

$$y(x) = cx^2,$$

$$y(t) = y(x(t)),$$

$$\frac{d}{dt}y(x(t)) = 2cx(t)V_0$$

$$\frac{d^2}{dt^2}y(x(t)) = \frac{d}{dt}(2cx(t)V_0) = 2cV_0 \frac{d}{dt}x(t) = 2cV_0^2.$$

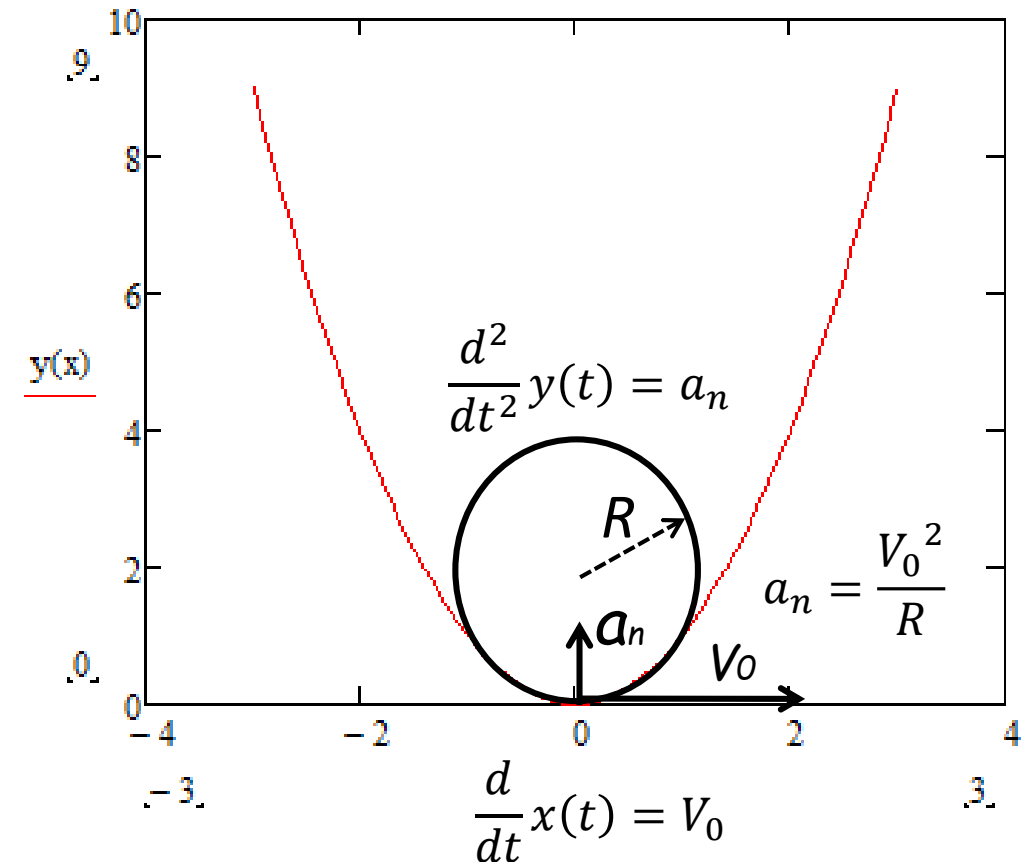
Use as a fact of circular motion. The normal acceleration is

$$a_n = \frac{V_0^2}{R}, \text{ where } R \text{ is the trajectory radius.}$$

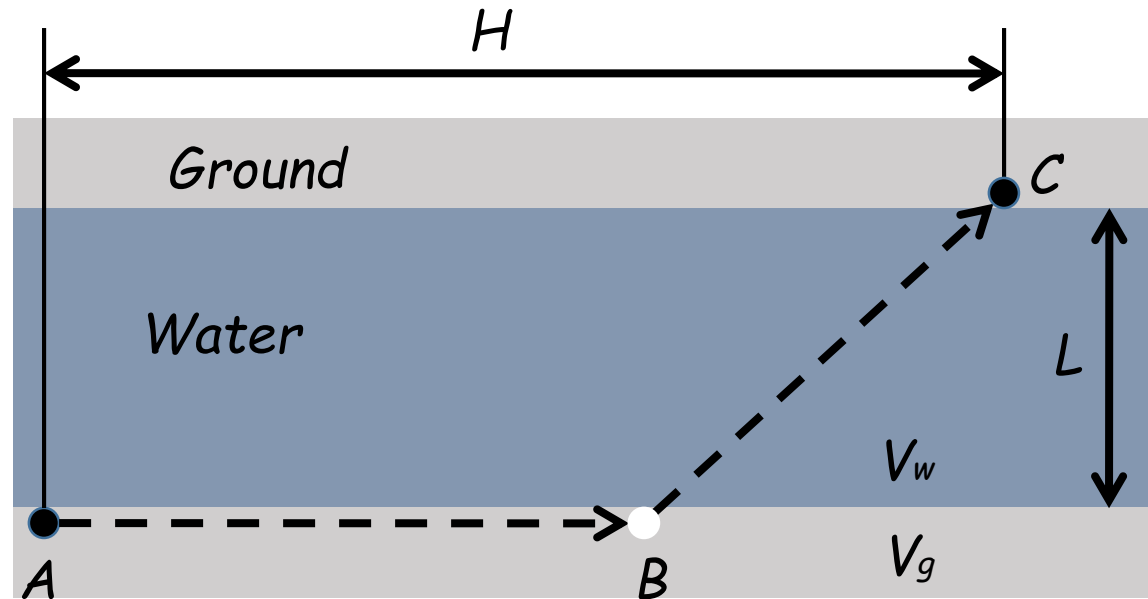
$$2cV_0^2 = \frac{V_0^2}{R}$$

$$\text{Answer: } a = a_n = 2cV_0^2, R = \frac{1}{2c}.$$

Physical solution



7. A traveler has to get at the final C position from the initial A position via the intermediate B point. His walking speed is 7 km/h and swimming speed is 2 km/h. Find the distance AB, which allows fastest travel from A to C. Take into account  $H = 200$  m,  $L = 45$  m.



7.

Let's construct a function  $t(AB)$

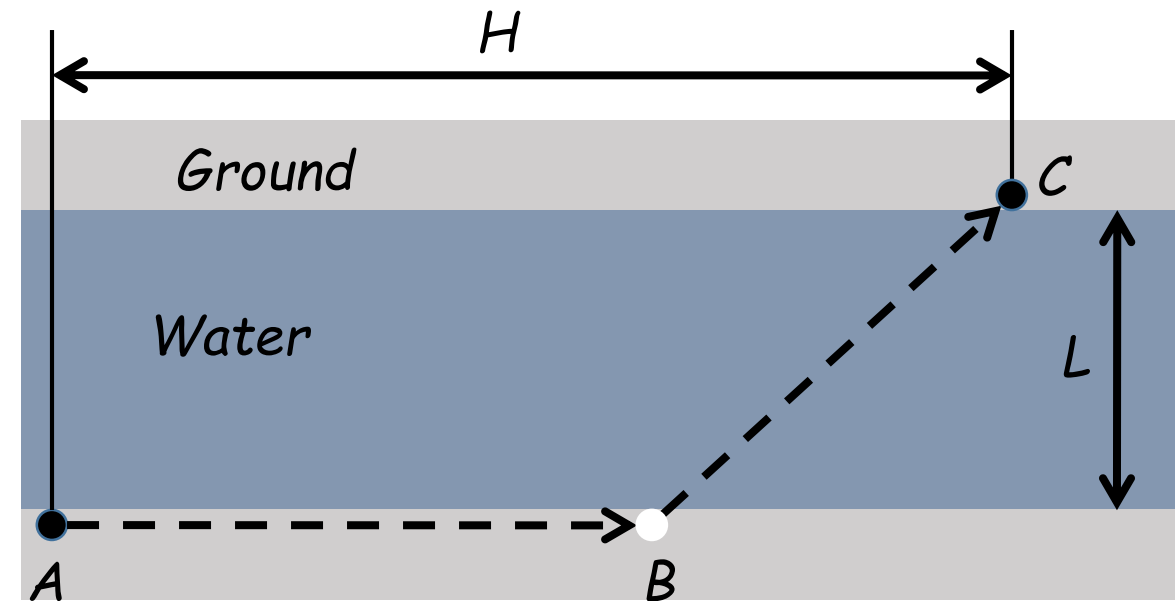
$$t(AB) = \frac{AB}{V_g} + \frac{\sqrt{(H - AB)^2 + L^2}}{V_w}$$

Finding extremum

$$\frac{d}{d(AB)} t(AB) = \frac{1}{V_g} + \frac{AB - H}{V_w \cdot \sqrt{(H - AB)^2 + L^2}}$$

$$\frac{1}{V_g} + \frac{AB - H}{V_w \cdot \sqrt{(H - AB)^2 + L^2}} = 0$$

**Answer:**  $AB = H - \frac{L \cdot V_w}{\sqrt{(V_g)^2 - (V_w)^2}} = 187 \text{ m}$



Solution 1

7.

Speed of BC-changing (shortening)

$$V_{BC} = V_g \cdot \cos(\alpha)$$

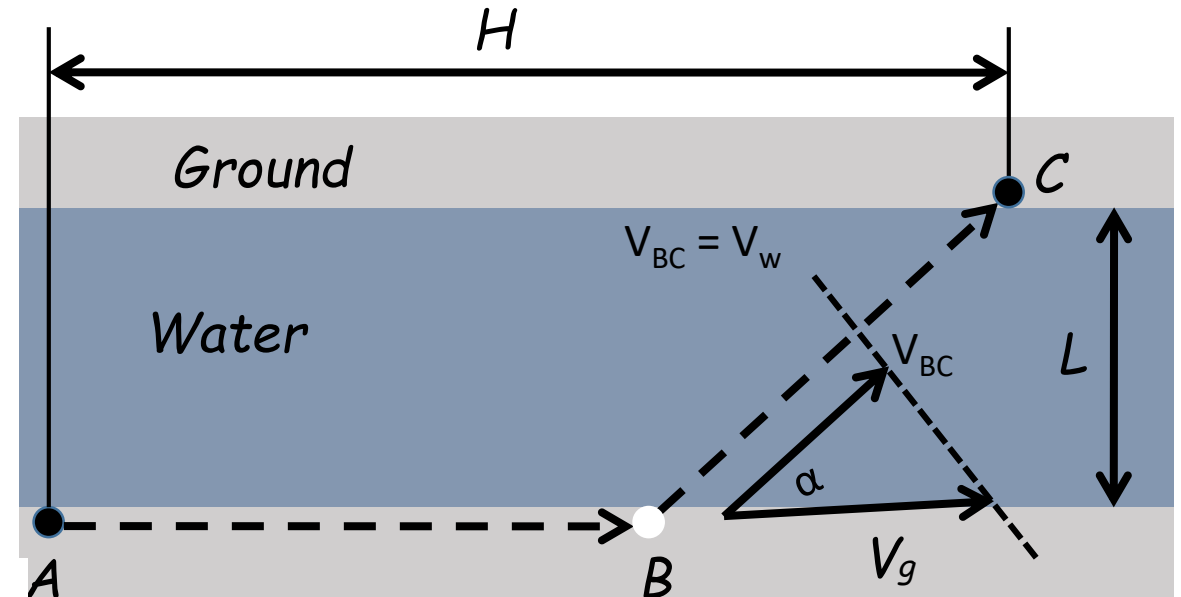
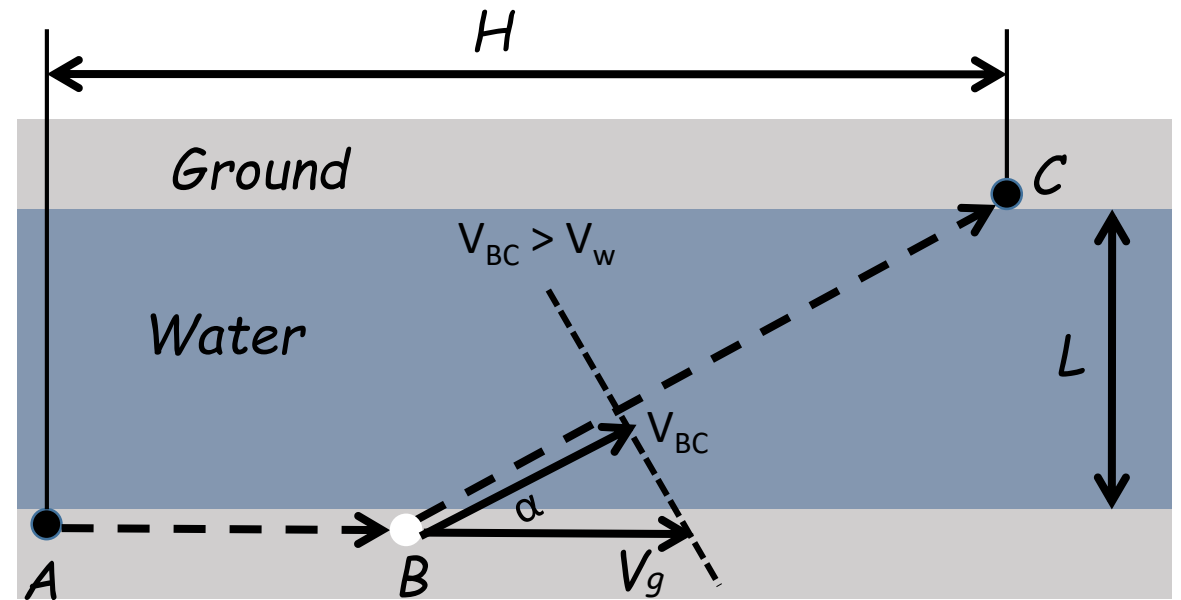
When should the traveler go to water?

$$V_g \cdot \cos(\alpha) = V_w$$

$$\cos(\alpha) = \frac{H - AB}{\sqrt{L^2 + (H - AB)^2}}$$

$$V_g \cdot \frac{H - AB}{\sqrt{L^2 + (H - AB)^2}} = V_w$$

$$\text{Answer: } AB = H - \frac{L \cdot V_w}{\sqrt{(V_g)^2 - (V_w)^2}} = 187 \text{ m}$$



Solution 2